NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

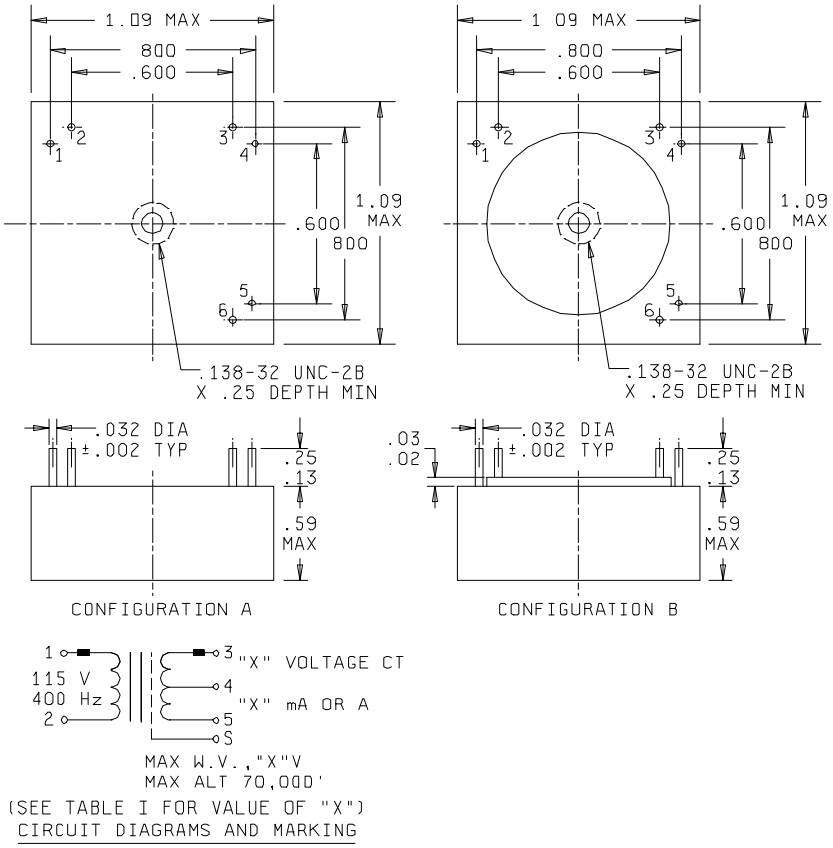
MIL-PRF-27/38B
7 February 1984
SUPERSEDING
MIL-T-27/38A
20 October 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 1.5 VOLT AMPERES, 400 HZ,
ELECTROSTATIC SHIELDED, TF5S03ZZ444 TO TF5S03ZZ487, INCLUSIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.138	3.51
.25	6.4
.59	15.0
.600	15.24
.75	19.1
.800	20.32
1.09	27.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

(B) denotes changes

FSC 5950

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight. 1.1 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Printed circuit pins - tinned brass.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure 1.25 times the maximum working voltage specified in table I.

Electrical characteristics.

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 3 mA rms.

Rated load. With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 70 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$ shall not exceed 18 percent.

Phase shift 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 17°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location See figure 1.

Ⓑ QUALITY ASSURANCE PROVISIONS:

Extent of qualification

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Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-T-27/37 through MIL-T-27/47 inclusive, all parts.

Qualification testing and approval to M27/38-44 shall be sufficient to grant qualification approval to M27/38-01 through M27/38-44.

Part number example:

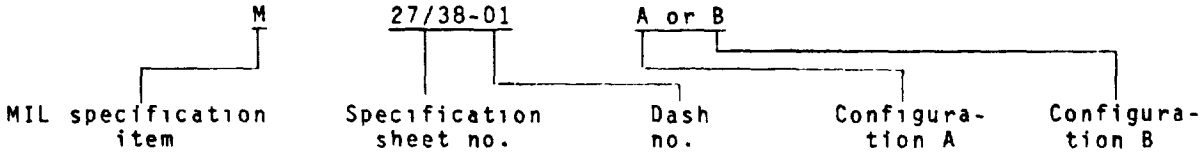


TABLE I. Electrical ratings.

Dash no.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	444	6	0.250	325
02 "	445	6.3	0.238	"
03 "	446	7	0.214	"
04 "	447	7.5	0.200	"
05 "	448	8	0.188	"
06 "	449	9	0.167	"
07 "	450	10	0.150	"
08 "	451	11	0.136	"
09 "	452	12	0.125	"
10 "	453	12.6	0.119	"
11 "	454	13	0.115	"
12 "	455	15	0.100	"
13 "	456	16	0.094	"
14 "	457	18	0.083	"
15 "	458	20	0.075	"
16 "	459	22	0.068	"
17 "	460	24	0.063	"
18 "	461	26	0.058	"
19 "	462	28	0.053	"
20 "	463	30	0.050	"
21 "	464	33	0.045	"
22 "	465	36	0.042	"
23 "	466	39	0.038	"
24 "	467	43	0.035	"
25 "	468	47	0.032	"
26 "	469	51	0.029	"
27 "	470	56	0.027	"
28 "	471	62	0.024	"
29 "	472	68	0.022	"
30 "	473	75	0.020	"
31 "	474	82	0.018	"
32 "	475	91	0.016	"
33 "	476	100	0.015	"
34 "	477	110	0.014	"
35 "	478	115	0.013	"
36 "	479	120	0.0125	"
37 "	480	130	0.0115	"
38 "	481	150	0.010	"
39 "	482	180	0.0085	"
40 "	483	200	0.0075	440
41 "	484	220	0.007	440
42 "	485	240	0.006	440
43 "	486	270	0.0055	580
44 "	487	300	0.005	580

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Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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